

DEFINITE INTEGRALS

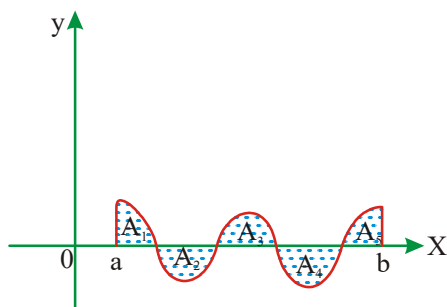


SYNOPSIS

- **Definition :** Let $f(x)$ be a function defined on $[a, b]$. If $\int f(x)dx = F(x)$, then $F(b) - F(a)$ is called the definite integral of $f(x)$ over $[a, b]$. It is denoted by $\int_a^b f(x)dx$. The real number 'a' is called the lower limit and the real number 'b' is called the upper limit.

$$\int_a^b f(x)dx = F(x) + c \Rightarrow \int_a^b f(x)dx = F(b) - F(a)$$

- **Geometrical Interpretation of Definite Integral :** Let $f(x)$ be a function defined on $[a, b]$. In general $\int_a^b f(x)dx$ represents an algebraic sum of areas of the region bounded by the curve $y = f(x)$, the x-axis and the ordinates $x = a$ and $x = b$. The area above the x-axis enters into this sum with positive sign while those below the x-axis enter with a negative sign.



Where A_1, A_2, A_3, A_4, A_5 are the areas of the shaded region.

- **General Properties of Definite Integral :**

$$1. \int_a^b f(x)dx = \int_a^b f(t)dt = \int_a^b f(y)dy$$

$$2. \int_a^a f(x)dx = 0$$

$$3. \int_a^b f(x)dx = -\int_b^a f(x)dx$$

$$4. \text{ If } a < c < b \text{ then } \int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$$

(where $C \in R$)

$$5. \int_a^b f(x)dx = \int_a^{c_1} f(x)dx + \int_{c_1}^{c_2} f(x)dx + \dots + \int_{c_n}^b f(x)dx$$

$$\text{where } a < c_1 < c_2 < \dots < c_n < b$$

$$6. \int_0^a f(x)dx = \int_0^a f(a-x)dx$$

$$7. \int_a^b f(x)dx = \int_a^b f(a+b-x)dx$$

$$8. \int_{-a}^a f(x)dx = \int_0^a [f(x) + f(-x)]dx$$

$$= \begin{cases} 2 \int_0^a f(x)dx & \text{if } f \text{ is even} \\ 0 & \text{if } f \text{ is odd} \end{cases}$$

$$9. \int_0^{2a} f(x)dx = \int_0^a [f(x) + f(2a-x)]dx$$

$$= \begin{cases} 2 \int_0^a f(x)dx & \text{if } f(2a-x) = f(x) \\ 0 & \text{if } f(2a-x) = -f(x) \end{cases}$$

$$10. \int_0^a f(x)dx = \int_0^{\frac{a}{2}} [f(x) + f(a-x)]dx$$

$$= \begin{cases} 2 \int_0^{\frac{a}{2}} f(x)dx, & \text{if } f(a-x) = f(x) \\ 0, & \text{if } f(a-x) = -f(x) \end{cases}$$

$$11. \int_a^b x f(x)dx = \frac{a+b}{2} \int_a^b f(x)dx$$

if $f(a+b-x) = f(x)$

$$12. \int_0^a x f(x) dx = \frac{a}{2} \int_0^a f(x) dx = a \cdot \int_0^{a/2} f(x) dx$$

$$\text{if } f(a-x) = f(x)$$

13. If $f(x)$ is an integrable function on $[a, b]$ and $g(x)$ is derivable on $[a, b]$ then

$$\int_a^b (f \circ g)(x) g'(x) dx = \int_{g(a)}^{g(b)} f(x) dx$$

14. If $f(x)$ is a periodic function with period T then

$$(i) \int_0^{nT} f(x) dx = n \int_0^T f(x) dx \text{ where } n \in \mathbb{N}$$

$$(ii) \int_a^{a+nT} f(x) dx = n \int_a^{a+T} f(x) dx = n \int_0^T f(x) dx$$

$$(iii) \int_{a+nT}^{b+nT} f(x) dx = \int_a^b f(x) dx \text{ where } n \in \mathbb{Z}$$

$$(iv) \int_{mT}^{nT} f(x) dx = (n-m) \int_0^T f(x) dx$$

$$\text{where } m, n \in \mathbb{Z}$$

W.E-1: Evaluate $\int_0^{100\pi} \sqrt{1+\cos 2x} dx$

Sol: $\sqrt{1+\cos 2x} = \sqrt{2\cos^2 x} = \sqrt{2} |\cos x|$
 $|\cos x|$ is a periodic function of fundamental period π

$$\therefore \int_0^{100\pi} \sqrt{1+\cos 2x} dx = \sqrt{2} \int_0^{100\pi} |\cos x| dx$$

$$= (\sqrt{2})(100) \int_0^{\pi} |\cos x| dx$$

$$= 100\sqrt{2} \left[\int_0^{\frac{\pi}{2}} \cos x dx + \int_{\frac{\pi}{2}}^{\pi} (-\cos x) dx \right]$$

$$= 100\sqrt{2} [1+1] = 200\sqrt{2}$$

W.E-2: Evaluate $\int_{-\pi}^{199\pi} \sqrt{\frac{1-\cos 2x}{2}} dx$

Sol: $\sqrt{\frac{1-\cos 2x}{2}} = |\sin x|$ is a periodic function of period π

$$\therefore \int_{-\pi}^{199\pi} \sqrt{\frac{1-\cos 2x}{2}} dx = \int_{-\pi}^{199\pi} |\sin x| dx$$

$$= [199 - (-1)] \int_0^{\pi} |\sin x| dx = 200 \int_0^{\pi} \sin x dx$$

$$= 200(2) = 400$$

➤ **Leibnitz Rule :**

15.

$$\frac{d}{dx} \left(\int_{\phi(x)}^{\psi(x)} f(t) dt \right) = f(\psi(x)) \frac{d}{dx} (\psi(x)) - f(\phi(x)) \frac{d}{dx} (\phi(x))$$

16. If $f(x) \geq 0$ in $[a, b]$ then $\int_a^b f(x) dx \geq 0$

17. If $f(x) \leq g(x)$ in $[a, b]$ then $\int_a^b f(x) dx \leq \int_a^b g(x) dx$

➤ **Approximation In Definite Integral :**

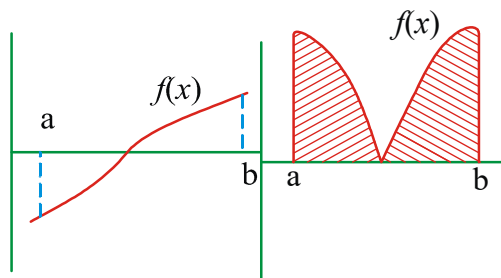
18. If $f_1(x) \leq f(x) \leq f_2(x) \forall x \in [a, b]$, then

$$\int_a^b f_1(x) dx \leq \int_a^b f(x) dx \leq \int_a^b f_2(x) dx$$

19. If absolute maximum and minimum value of $f(x)$, when $x \in [a, b]$ is M and m respectively, then

$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$$

21. $\int_a^b f(x) dx \leq \int_a^b |f(x)| dx$



22. $\left| \int_a^b f(x) g(x) dx \right| \leq \sqrt{\int_a^b f^2(x) dx} \sqrt{\int_a^b g^2(x) dx}$

➤ **Mean value of a function over an interval:**

let $f(x)$ be a continuous function define on $[a, b]$
then there exist a point $C \in (a, b)$ such that

$$\int_a^b f(x) dx = f(c)(b-a) \text{ then}$$

$f(c) = \frac{1}{b-a} \int_a^b f(x) dx$ is called the mean value
of $f(x)$ over $[a, b]$

➤ **Reduction Formulae :**

$$23. \text{ If } I_n = \int_0^{\pi/2} \sin^n x dx = \int_0^{\pi/2} \cos^n x dx, \text{ then } I_n = \frac{n-1}{n} I_{n-2}$$

where $n \in \mathbb{Z}^+$

$$I_n = \left(\frac{n-1}{n} \right) \left(\frac{n-3}{n-2} \right) \dots \frac{1}{2} \cdot \frac{\pi}{2}, \text{ if } n \text{ is even}$$

$$= \left(\frac{n-1}{n} \right) \left(\frac{n-3}{n-2} \right) \dots \frac{2}{3} \text{ if } n \text{ is odd}$$

$$24. \text{ If } I_n = \int_0^{\pi/4} \tan^n x dx \text{ (} n > 1 \text{) then}$$

$$I_n + I_{n-2} = \frac{1}{n-1} \text{ and hence}$$

$$I_n = \frac{1}{n-1} - \frac{1}{n-3} + \frac{1}{n-5} - \frac{1}{n-7} \dots \dots \dots I_0 \text{ or } I_1$$

according as n is even or odd.

$$\text{Here } I_0 = \frac{\pi}{4}, I_1 = \frac{1}{2} \log 2.$$

$$25. \text{ a) } \int_0^{\pi/4} (\tan^n x + \tan^{n-2} x) dx = \frac{1}{n-1}$$

$$\text{b) } \int_{\pi/4}^{\pi/2} [\cot^n x + \cot^{n-2} x] dx = \frac{1}{n-1}$$

$$26. \text{ If } I_n = \int_0^{\pi/4} \sec^n x dx \text{ (} n \in \mathbb{Z}^+ \text{) then}$$

$$I_n = \frac{(\sqrt{2})^{n-2}}{n-1} + \frac{n-2}{n-1} I_{n-2}.$$

$$27. \int_{\pi/4}^{\pi/2} \sec^n x dx = \frac{(\sqrt{2})^{n-2}}{n-1} + \frac{n-2}{n-1} \int_{\pi/4}^{\pi/2} \sec^{n-2} x dx$$

➤ **Walli's Formulae :**

$$28. \text{ If } I_{m,n} = \int_0^{\pi/2} \sin^m x \cos^n x dx \text{ then}$$

$$\text{a) } I_{m,n} = \frac{m-1}{m+n} I_{m-2,n} \cdot \left(m, n \in \mathbb{Z}^+ \right)$$

$$\text{b) } I_{m,n} = \frac{n-1}{m+n} I_{m,n-2} \cdot \left(m, n \in \mathbb{Z}^+ \right)$$

$$30. \int_0^{\pi/2} \sin^m x \cos^n x dx = \frac{[(m-1)(m-3)(m-5)\dots][(n-1)(n-3)\dots]}{(m+n)(m+n-2)(m+n-4)\dots} \times k$$

$$\text{If } m \text{ and } n \text{ are both even then } k = \frac{\pi}{2},$$

$$\text{otherwise } k = 1$$

$$31. I_n = \int_0^{\pi/2} x^n \sin x dx = n \left(\frac{\pi}{2} \right)^{n-1} - n(n-1) I_{n-2}$$

$$32. I_n = \int_0^{\pi/2} x^n \cos x dx = \left(\frac{\pi}{2} \right)^n - n(n-1) I_{n-2}$$

➤ **Standard Results :**

$$1. \int_a^b \frac{|x|}{x} dx = |b| - |a| = \begin{cases} b-a & \text{if } 0 < a < b \\ b+a & \text{if } a < 0 < b \\ a-b & \text{if } a < b < 0 \end{cases}$$

$$2. \int_0^{\pi} \sin^m x \cos^n x dx$$

$$= \begin{cases} \frac{\pi}{2} \int_0^{\pi/2} \sin^m x \cos^n x dx & \text{if } n \text{ is even} \\ 0 & \text{if } n \text{ is odd} \end{cases}$$

$$3. \int_0^{2\pi} \sin^m x \cos^n x dx = \begin{cases} 4 \int_0^{\frac{\pi}{2}} \sin^m x \cos^n x dx \\ 0 \text{ otherwise} \end{cases}$$

if both n and m are even

$$4. \text{ i) } \int_0^{\frac{\pi}{4}} \log(1 + \tan \theta) d\theta = \frac{\pi}{8} \log 2$$

$$\text{ii) } \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \log(1 + \cot \theta) d\theta = \frac{\pi}{8} \log 2$$

W.E-3 : $\int_0^{\frac{\pi}{2}} \log(\sin x) dx = -\frac{\pi}{2} \log 2$

Sol: $I = \int_0^{\frac{\pi}{2}} \log(\sin x) dx \quad \text{---1}$

$$\int_0^{\frac{\pi}{2}} \log\left(\sin\left(\frac{\pi}{2} - x\right)\right) dx; I = \int_0^{\frac{\pi}{2}} \log(\cos x) dx \quad \text{---2}$$

(1) + (2) $\Rightarrow$

$$\therefore 2I = \int_0^{\frac{\pi}{2}} (\log(\sin x) + \log(\cos x)) dx$$

$$= \int_0^{\frac{\pi}{2}} \log\left(\frac{2 \sin x \cos x}{2}\right) dx = \int_0^{\frac{\pi}{2}} \log\left(\frac{\sin 2x}{2}\right) dx$$

$$= \int_0^{\frac{\pi}{2}} \log(\sin 2x) dx - \int_0^{\frac{\pi}{2}} \log 2 dx \quad \text{--- 3}$$

where $\int_0^{\frac{\pi}{2}} \log(\sin 2x) dx$ put $2x = t$

$$= 1/2 \int_0^{\pi} \log(\sin t) dt = \frac{1}{2} \cdot 2 \cdot \int_0^{\frac{\pi}{2}} \log(\sin t) dt$$

$$= \int_0^{\frac{\pi}{2}} \log(\sin t) dt = I \quad \text{--- 4}$$

(From 3 & 4) $2I = I - \frac{\pi}{2} \log 2; \quad I = -\frac{\pi}{2} \log 2$

$$5. \text{ i) } \int_0^{\frac{\pi}{2}} \log(\sin x) dx = \int_0^{\frac{\pi}{2}} \log(\cos x) dx = -\frac{\pi}{2} \log 2$$

$$\text{ii) } \int_0^{\frac{\pi}{2}} \log(\tan x) dx = \int_0^{\frac{\pi}{2}} \log(\cot x) dx = 0$$

$$6. \text{ (i) } \int_0^{\frac{\pi}{2}} \frac{a \cos x + b \sin x}{\cos x + \sin x} dx = \frac{\pi}{4} (a + b)$$

$$\text{(ii) } \int_0^{\frac{\pi}{2}} \frac{a \tan x + b \cot x}{\tan x + \cot x} dx = \frac{\pi}{4} (a + b)$$

$$\text{(iii) } \int_0^{\frac{\pi}{2}} \frac{a \sec x + b \csc x}{\sec x + \csc x} dx = \frac{\pi}{4} (a + b)$$

$$7. \text{ (i) } \int_0^{\frac{\pi}{2}} \frac{f(\sin x)}{f(\sin x) + f(\cos x)} dx = \frac{\pi}{4}$$

$$\text{(ii) } \int_0^{\frac{\pi}{2}} \frac{f(\tan x)}{f(\tan x) + f(\cot x)} dx = \frac{\pi}{4}$$

$$\text{(iii) } \int_0^{\frac{\pi}{2}} \frac{f(\sec x)}{f(\sec x) + f(\csc x)} dx = \frac{\pi}{4}$$

$$8. \int_0^{\frac{\pi}{2}} \frac{1}{\sin x + \cos x} dx = \sqrt{2} \log(\sqrt{2} + 1)$$

$$9. \int_0^{\frac{\pi}{2}} \frac{1}{a^2 \sin^2 x + b^2 \cos^2 x} dx = \frac{\pi}{2ab}$$

$$\int_0^{\pi} \frac{x}{a^2 \sin^2 x + b^2 \cos^2 x} dx = \frac{\pi^2}{2ab}$$

$$10. \int_0^{\pi} \frac{x \tan x}{\sec x + \cos x} dx = \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = \frac{\pi^2}{4}$$

$$11. \int_0^{\pi} \frac{x \tan x}{\sec x + \tan x} dx = \int_0^{\pi} \frac{x \sin x}{1 + \sin x} dx = \frac{\pi(\pi - 2)}{2}$$

12. (i) If $a > 0$, $\int_0^{\infty} e^{-ax} \cos bx \, dx = \frac{a}{a^2 + b^2}$

(ii) If $a > 0$, $\int_0^{\infty} e^{-ax} \sin bx \, dx = \frac{b}{a^2 + b^2}$

13. $\int_0^{\infty} \frac{1}{(x + \sqrt{x^2 + 1})^n} dx =$

$\int_0^{\pi/2} \frac{\sec^2 x}{(\sec x + \tan x)^n} dx = \frac{n}{n^2 - 1}$

14. (i) $\int_a^b \sqrt{(x-a)(b-x)} \, dx = \frac{\pi}{8} (b-a)^2$

(ii) $\int_a^b \frac{1}{\sqrt{(x-a)(b-x)}} \, dx = \pi$

(iii) $\int_a^b \frac{1}{x\sqrt{(x-a)(b-x)}} \, dx = \frac{\pi}{\sqrt{ab}}$

(iv) $\int_a^b \sqrt{\frac{x-a}{b-x}} \, dx = \int_a^b \sqrt{\frac{b-x}{x-a}} \, dx = \frac{\pi}{2} (b-a)$

(v) $\int_0^a \sqrt{\frac{a+x}{a-x}} \, dx = a \left(\frac{\pi+2}{2} \right)$

(vi) $\int_0^a \sqrt{\frac{a-x}{a+x}} \, dx = a \left(\frac{\pi-2}{2} \right)$

15. (i) $\int_0^n [x] dx = \frac{n(n-1)}{2} = \sum_{r=1}^n \left(\int_{r-1}^r [x] dx \right) \forall n \in N$

(where $[]$ is g.i.f)

(ii) $\int_0^n \{x\} \, dx = \frac{n}{2}, \forall n \in N$

(where $\{x\}$ is fractional part of x)

(iii) $\int_0^k \{x\} \, dx = \int_0^1 \sqrt{x} \, dx + \int_1^k \sqrt{x} - 1 \, dx$ if $k \in I$ and perfect square

(iv) $\int_0^{n^2} [\sqrt{x}] \, dx = \frac{n(n-1)(4n+1)}{6}, \forall n \in N$

(where $[]$ is g.i.f)

W.E-4: Evaluate $\int_0^n [x] dx \, (n \in N)$ where $[]$

denotes the G I F

Sol:- We know that $k \leq x < k+1 \Rightarrow [x] = k, k \in Z$

$$\begin{aligned} \therefore \int_0^n [x] \, dx &= \int_0^1 [x] \, dx + \int_1^2 [x] \, dx + \dots + \int_{n-1}^n [x] \, dx \\ &= \int_0^1 0 \, dx + \int_1^2 1 \, dx + \int_2^3 2 \, dx + \dots + \int_{n-1}^n (n-1) \, dx \\ &= 0 + 1(2-1) + 2(3-2) + \dots + (n-1)[n-(n-1)] \\ &= 1 + 2 + 3 + \dots + (n-1) = \frac{n(n-1)}{2} \end{aligned}$$

W.E-5: $\int_0^2 [x^2] \, dx$ where $[]$ denotes the GIF

Sol: $\int_0^1 0 \, dx + \int_1^{\sqrt{2}} 1 \, dx + \int_{\sqrt{2}}^{\sqrt{3}} 2 \, dx + \int_{\sqrt{3}}^2 3 \, dx =$
 $= 0 + 1(\sqrt{2}-1) + 2(\sqrt{3}-\sqrt{2}) + 3(2-\sqrt{3})$
 $= 5 - \sqrt{2} - \sqrt{3}$

W.E-6: Evaluate $\int_0^{100} [\tan^{-1} x] \, dx$

Sol: The range of $\tan^{-1} x$ is $\left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$

$\therefore$ When $x \in [0, 100], 0 \leq x < \tan 1 \Rightarrow [\tan^{-1} x] = 0$

$\tan 1 \leq x < \infty \Rightarrow [\tan^{-1} x] = 1$

$\int_0^{100} [\tan^{-1} x] \, dx = \int_0^{\tan 1} 0 \, dx + \int_{\tan 1}^{100} 1 \, dx = 100 - \tan 1$

16. $\int_0^{\pi/2} \cos^{n-2} x \sin nx \, dx = \frac{1}{n-1}$ if $n > 1$

17. $\int_0^1 x^m (\log x)^n \, dx = \frac{(-1)^n n!}{(m+1)^{n+1}}$

or $\int_0^1 x^{\alpha-1} (\log x)^n \, dx = \frac{(-1)^n n!}{\alpha^{n+1}}$

$$18. (i) \int_0^1 x^m (1-x)^n dx = \int_0^1 x^n (1-x)^m dx$$

$$= \frac{m!n!}{(m+n+1)!} \forall m, n \in N$$

$$(ii) \int_0^1 x^{m-1} (1-x)^{n-1} dx = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx$$

$$= \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}} dx$$

$$19. \int_0^1 (1+x)^m (1-x)^n dx = \frac{2^{m+n+1} m!n!}{(m+n+1)!}$$

$$20. \int_{-\infty}^\infty \frac{x^c}{c^x} dx = \frac{\sqrt{c+1}}{(\log c)^{c+1}}$$

$$21. \int_0^{2n\pi} [\sin x + \cos x] dx = \int_0^{2n\pi} \left[\sqrt{2} \sin \left(x + \frac{\pi}{4} \right) \right] dx$$

$$= n \int_0^{2\pi} \left[\sqrt{2} \sin \left(x + \frac{\pi}{4} \right) \right] dx = -n\pi$$

(where [] is g.i.f)

$$22. (i) \int_0^{n\pi} \sin \left[\frac{2x}{\pi} \right] dx = \frac{\frac{\pi}{2} \sin(n) \sin \left(\frac{2n-1}{2} \right)}{\sin \left(\frac{1}{2} \right)}$$

(where [] is g.i.f)

$$(ii) \int_0^\infty [f(x)] dx = \log \left(\frac{n^n}{n!} \right) \text{ where}$$

$$f(x) = \frac{n}{e^x} = ne^{-x}, n \in N \text{ (where [] is g.i.f)}$$

➤ **Integration as limit of a sum :**

$$22. \int_0^1 f(x) dx = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n f \left(\frac{r-1}{n} \right) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{n-1} f \left(\frac{r}{n} \right)$$

replace $\frac{r}{n}$ by x , $\frac{1}{n}$ by dx and $\lim_{n \rightarrow \infty} \sum$ by $\int$

$$23. \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{kn} f \left(\frac{r-1}{n} \right) = \int_0^k f(x) dx$$



C.U.Q

1. If $f(x)$ is integrable on $[0, a]$ then

$$\int_0^a \frac{f(x)}{f(x) + f(a-x)} dx =$$

- 1) 0 2) 1 3) a 4) $\frac{a}{2}$

$$2. \int_0^{\pi/2} \frac{a \cos x + b \sin x}{\cos x + \sin x} dx =$$

- 1) $\pi(a+b)$ 2) $\frac{\pi}{2}(a+b)$
3) $\frac{\pi}{4}(a+b)$ 4) πab

3. Value of the integral $I = \int_0^1 x(1-x)^n dx =$

- 1) $\frac{1}{n+2}$ 2) $\frac{1}{n+1} - \frac{1}{n+2}$
3) $\frac{1}{n+1} + \frac{1}{n+2}$ 4) $\frac{1}{n+1}$

4. If $f(a+b-x)=f(x)$ then $\int_a^b x \cdot f(x) dx =$

- 1) $\frac{a+b}{2} \int_a^b f(b-x) dx$ 2) $\frac{a+b}{2} \int_a^b f(x) dx$
3) $\frac{b-a}{2} \int_a^b f(x) dx$ 4) $(a+b) \int_a^b f(x) dx$

$$5. \int_0^\pi x f(\sin x) dx =$$

- 1) 2π 2) $\pi \int_0^{\pi/2} f(\cos x) dx$
3) $\pi \int_0^\pi f(\cos x) dx$ 4) $\pi \int_0^\pi f(\sin x) dx$

6. If $f(x)$ and $g(x)$ are any two continuous

functions $\forall x \in \mathbb{R}$ then $\int_{-a}^a \frac{g(x) - g(-x)}{f(x) + f(-x)} dx$

- 1) 0 2) 4 3) 6 4) 3

7. If $\int_0^a x^m (a-x)^n dx = k$ then $\int_0^a x^n (a-x)^m dx =$

- 1) k 2) $-k$ 3) $k/2$ 4) $k/3$

8. $\int_0^{2a} \frac{f(x) dx}{f(x) + f(2a-x)}$

- 1) a 2) $a/2$ 3) 0 4) $2a$

9. $\int_a^b f(a+b-x) dx = \int_a^b f(x) dx$ is true

- 1) For all values of a, b 2) For $a = 0$ only

- 3) For all values of a only

- 4) For all values of b only

10. $\int_0^\pi \frac{x dx}{1 + \sin x} =$

- 1) $\frac{\pi}{6}$ 2) π 3) $\frac{\pi}{3}$

- 4) Cannot be valued

11. $\int_{-\pi/2}^{\pi/2} \sin^7 x dx =$

- 1) 0 2) 1 3) 2 4) 3

12. $\int_0^\pi \log(1 + \cos x) dx =$

- 1) $-\frac{\pi}{2} \log 2$ 2) $-\frac{\pi}{3}$

- 3) $-\pi \log 2$ 4) $-2\pi \log 2$

13. Evaluate $\int_0^1 e^{2-3x} dx$

- 1) $e^2 - e$ 2) $\frac{1}{3}(e^2 - e)$
3) $\frac{1}{3}\left(e^2 - \frac{1}{e}\right)$ 4) $\frac{1}{2}\left(e^2 - \frac{1}{e}\right)$

14. If $f(x)$ is even function then

$\int_{-\pi}^{\pi} f(x) \cdot \sin nx dx =$

- 1) 0 2) $\int_0^\pi f(x) \cdot \sin nx dx$

3) $4 \int_0^{\pi/2} f(x) \cdot \sin nx dx$

4) $2 \int_0^{\pi/2} f(x) \cdot \sin nx dx$

15. $\int_{-1}^1 (ax^3 + bx) dx = 0$ for

- 1) for all values of a and b

- 2) $a > 0$ and $b > 0$ only

- 3) $a > 0$ and $b < 0$ only

- 4) $a < 0$ and $b < 0$ only

16.

$\lim_{n \rightarrow \infty} \left[\frac{\sqrt{n^2 - 1^2}}{n^2} + \frac{\sqrt{n^2 - 2^2}}{n^2} + \frac{\sqrt{n^2 - 3^2}}{n^2} + \dots n \text{ terms} \right] =$

- 1) $\frac{\pi}{4}$ 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{3}$ 4) $\frac{2\pi}{4}$

17. $\int_0^{\pi/2} \frac{\sec x}{\sec x + \csc x} dx =$

- 1) $\frac{\pi}{3}$ 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{4}$ 4) $\frac{\pi}{8}$

18. $\int_0^{\pi/2} (\cos x - \sin x) e^x dx =$

- 1) 1 2) 0 3) -1 4) $\frac{\pi}{2}$

19. $\int_a^b \frac{f(x)}{f(x) + f(a+b-x)} dx =$

- 1) $a-b$ 2) $b-a$ 3) $\frac{b-a}{2}$ 4) $\frac{a-b}{2}$

20. $\int_0^n [x] dx =$

- 1) n 2) $\frac{n(n+1)}{2}$ 3) $\frac{n(n-1)}{2}$ 4) 0

21. The integral $\int_0^a \frac{g(x)}{f(x) + f(a-x)} dx$ vanishes, if

- 1) $g(x)$ is odd

- 2) $f(x) = f(a-x)$

- 3) $g(x) = -g(a-x)$

- 4) $f(a-x) = -g(x)$

22. If $\frac{d}{dx} f(x) = g(x)$ then $\int_a^b f(x) g(x) dx =$

- 1) $\frac{f(b)-f(a)}{2}$ 2) $\frac{f(a)-f(b)}{2}$
 3) $\frac{f^2(b)-f^2(a)}{2}$ 4) $\frac{f^2(a)-f^2(b)}{2}$

23. $\int_0^{\pi/2} \log \cot x dx =$

- 1) 1 2) π 3) 0 4) $\frac{\pi}{2}$

24. If f is a continuous periodic function on $\mathbb{R}$,

then $\int_{mT}^{nT} f(x) dx =$

- 1) $\int_0^T f(x) dx$ 2) $(n+m) \int_0^T f(x) dx$
 3) $(n-m) \int_0^T f(x) dx$ 4) $\frac{n+m}{2} \int_0^T f(x) dx$

25. If $a < b < 0$ then $\int_a^b \frac{|x|}{x} dx =$

- 1) $a-b$ 2) $b-a$ 3) $a+b$ 4) 0

26. If $a < 0 < b$, then $\int_a^b \frac{|x|}{x} dx =$

- 1) $b-a$ 2) $a-b$ 3) $a+b$ 4) 0

27. If $I_n = \int_0^{\pi/4} \tan^n x dx$ then $\frac{1}{I_3 + I_5} =$

- 1) $\frac{1}{4}$ 2) $\frac{1}{2}$ 3) $\frac{1}{8}$ 4) 4

28. $\int_0^{\pi/2} \frac{\sec^2 x dx}{(\sec x + \tan x)^n} =$ ($n > 2$)

- 1) $\frac{1}{n^2-1}$ 2) $\frac{n}{n^2-1}$ 3) $\frac{n}{n^2+1}$ 4) $\frac{2}{n^2-1}$

29. $\int_0^{2\pi} e^{ax} \cos bx dx$; $a, b \in \mathbb{Z} =$

- 1) $\frac{a}{a^2+b^2}(e^{2a\pi}-1)$ 2) $\frac{b}{a^2+b^2}(e^{2a\pi}-1)$
 3) $\frac{a}{a^2+b^2}$ 4) $\frac{b}{a^2+b^2}$

C.U.Q- KEY

- 1) 4 2) 3 3) 2 4) 2 5) 2 6) 1
 7) 1 8) 1 9) 1 10) 2 11) 1 12) 3
 13) 3 14) 1 15) 1 16) 1 17) 3 18) 3
 19) 3 20) 3 21) 3 22) 3 23) 3 24) 3
 25) 1 26) 3 27) 4 28) 2 29) 1

C.U.Q - HINTS

1. $\int_0^a \frac{f(x)}{f(x)+f(a-x)} dx = \frac{a}{2}$

2. $\int_0^{\frac{\pi}{2}} \frac{a \cos x + b \sin x}{\cos x + \sin x} dx = \frac{\pi}{4}(a+b)$

3. using formula $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

4. Using formula $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$

5. $I = \int_0^{\pi} x f(\sin x) dx$;

$I = \int_0^{\pi} (\pi-x) f(\sin x) dx$;

$I = \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx$ put $\frac{\pi}{2} - x = t$

6. Given function is odd

7. Using formula $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

8. Using formula $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

9. For all values of a, b

10. $I = \int_0^{\pi} \frac{x dx}{1 + \sin x}$

Use Property $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

11. $I = \int_{-\pi/2}^{\pi/2} \sin^7 x dx$; $f(-x) = -f(x)$
 Odd function $I = 0$

$$12. I = \int_0^{\pi} \log(1 + \cos x) dx$$

$$I = \int_0^{\pi} \log[(1 + \cos(\pi - x))] dx$$

$$= \int_0^{\pi} \log[1 - \cos x] dx$$

$$2I = \int_0^{\pi} \log[\sin^2 x] dx = -2\pi \log 2$$

$$13. I = \int_0^1 e^{2-3x} dx = e^2 \int_0^1 e^{-3x} dx$$

$$14. \text{In } [-\pi, \pi], f(x) \text{ is even and } \sin nx \text{ is odd} \\ \Rightarrow f(x) \cdot \sin nx \text{ is odd function.}$$

$$15. ax^3 + bx \text{ is an odd function for all values of } a \text{ and } b.$$

$$16. \text{Given series} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \sqrt{1 - \left(\frac{r}{n}\right)^2}$$

$$17. \int_0^{\pi/2} \frac{f(\sec x)}{f(\sec x) + f(\cos ec x)} dx = \frac{\pi}{4}.$$

$$18. \int [f(x) + f'(x)] e^x dx = e^x \cdot f(x).$$

$$20. \int_0^n [x] dx = 0 + 1 + 2 + \dots + (n-1) = \frac{n(n-1)}{2}$$

$$21. g(x) = -g(a-x)$$

$$\text{Clearly } I \text{ vanishes if } g(a-x) = -g(x).$$

$$22. \int_a^b f(x) f'(x) dx = \left[f^2(x) \right]_a^b$$

$$23. \int_0^a f(x) dx = \int_0^a f(a-x) dx.$$

$$25. \int \frac{|x|}{x} dx = |x|$$

$$26. \int_a^b \frac{|x|}{x} dx = \int_a^0 \frac{|x|}{x} dx + \int_0^b \frac{|x|}{x} dx$$

$$27. I_n = \int_0^{\pi/4} \tan^n x dx \text{ then } I_n + I_{n+2} = \frac{1}{n+1}.$$

$$28. \frac{n}{n^2 - 1}$$

$$29. \int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2 + b^2} [a \cos bx + b \sin bx]$$



LEVEL-I (C.W)

PROBLEMS BASED ON BASIC FORMULAE

$$1. \int_0^4 x\sqrt{x} dx =$$

- 1) 12.4 2) 8.4 3) 8.8 4) 12.8

$$2. \frac{1}{\pi} \int_{-2}^2 \frac{1}{4+x^2} dx =$$

- 1) 0 2) $\frac{1}{4}$ 3) $\frac{\pi}{4}$ 4) $\frac{1}{2}$

$$3. \text{If } \int_{\sqrt{2}}^x \frac{dx}{x\sqrt{x^2-1}} = \frac{\pi}{12}, \text{ then } x =$$

- 1) 1 2) $\frac{1}{2}$ 3) 2 4) -2

$$4. \text{If } \int_0^k \frac{dx}{2+8x^2} = \frac{\pi}{16} \text{ then } k =$$

1. 1 2. $\frac{1}{2}$ 3. $\pi/2$ 4. 2

$$5. \text{If } \int_0^{40} \frac{dx}{2x+1} = \log a \text{ then } a$$

- 1) 3 2) 9 3) 81 4) 40

$$6. \int_0^{\pi/2} \frac{\sin^2 x}{(1 + \cos x)^2} dx =$$

1. $\pi/2$ 2. $2 - \pi/2$
3. $\pi/2 - 2$ 4. $2 + \pi/2$

$$7. \int_{1/2}^1 \frac{1}{\sqrt{x-x^2}} dx =$$

1. $\frac{1}{2}$ 2. $\pi/4$ 3. $\pi/2$ 4. π

$$8. \text{Value of } I = \int_0^{\pi/2} \frac{(\sin x + \cos x)^2}{\sqrt{1 + \sin 2x}} dx \text{ is}$$

- 1) 0 2) 1 3) 2 4) 4

$$9. \int_0^{\pi/2} \sqrt{1 + \sin x} dx =$$

- 1) 1 2) $1/2$ 3) 2 4) 3

10. $\int_0^{\log 2} \cosh 2x dx =$
1) 15/16 2) -15 3) 16/17 4) 17/18
11. $\int_{\pi/2}^{3\pi/2} \frac{1}{1+\cos x} dx =$
1) 2 2) -2 3) 1/2 4) -1/2
12. $\int_0^{16} \frac{dx}{\sqrt{x+9}-\sqrt{x}} =$
1) 10 2) 12 3) 14 4) 16
13. If $0 < a < c, 0 < b < c$ then $\int_0^{\infty} \frac{a^x - b^x}{c^x} dx =$
1) $\log \frac{b}{c} - \log \frac{a}{c}$ 2) $\frac{\log a - \log b}{\log c}$
3) $\frac{1}{\log b/c} - \frac{1}{\log a/c}$ 4) $\log \frac{a}{c} - \log \frac{b}{c}$
14. If $f(x) = \begin{vmatrix} x & \cos x & e^{x^2} \\ \sin x & x^2 & \sec x \\ \tan x & x^4 & 2x^2 \end{vmatrix}$ then
 $\int_{-\pi/2}^{\pi/2} f(x) dx =$
1) 0 2) 1 3) 2 4) 3

PROBLEMS BASED ON SUBSTITUTIONS

15. $\int_0^1 \frac{dx}{e^x + e^{-x}} =$
1) $\tan^{-1} e$ 2) $\frac{\pi}{4}$ 3) $\tan^{-1} e - \frac{\pi}{4}$ 4) $\tan^{-1} e + \frac{\pi}{4}$
16. $\int_0^1 \frac{4x^3}{\sqrt{1-x^8}} dx =$
1) π 2) $-\pi$ 3) $\frac{\pi}{2}$ 4) $-\frac{\pi}{2}$
17. If $\int_0^k \frac{\cos x}{1+\sin^2 x} dx = \frac{\pi}{4}$ then $k =$
1) 1 2) $\pi/4$ 3) $\pi/2$ 4) $\pi/6$

18. $\int_0^{\pi/4} \frac{\sqrt{\tan x}}{\sin x \cos x} dx =$
1) 1 2) 2 3) 0 4) 4
19. $\int_0^{\pi/2} \cos^5 x \cdot \sin 2x dx =$
1) 2/7 2) 1/7 3) -1/7 4) 3/7
20. $\int_1^2 x^{2x} [1 + \log x] dx =$
1) $\frac{9}{2}$ 2) $\frac{11}{2}$ 3) $\frac{13}{2}$ 4) $\frac{15}{2}$
21. $\int_n^{n+1} f(x) dx = n^2 + n$ then $\int_{-1}^1 f(x) dx =$
1) 0 2) -2 3) 3 4) 4
22. $\int_{\frac{1}{\pi}}^{\frac{2}{\pi}} \frac{\sin(1/x)}{x^2} dx$
1) $\frac{\pi}{2}$ 2) π 3) 1 4) 0
23. $\int_1^{32} \frac{dx}{x^{1/5} \sqrt{1+x^{4/5}}}$
1) $\frac{2}{5}(\sqrt{17} + \sqrt{2})$ 2) $\frac{2}{5}(\sqrt{17} - \sqrt{2})$
3) $\frac{5}{2}(\sqrt{17} - \sqrt{2})$ 4) $\frac{5}{2}(\sqrt{17} + \sqrt{2})$
24. If $I_n = \int_0^{\infty} e^{-x} x^{n-1} dx$, then $\int_0^{\infty} e^{-\lambda x} x^{n-1} dx$ is equal to
1) λI_n 2) $\frac{1}{\lambda} I_n$ 3) $\frac{I_n}{\lambda^n}$ 4) $\lambda^n I_n$
25. If $\int_{-1}^4 f(x) dx = 4$ and $\int_2^4 (3 - f(x)) dx = 7$ then
 $\int_{-1}^2 f(x) dx =$
1) -2 2) 3 3) 5 4) 8

26. If $\int_0^{10} f(x) dx = 5$ then $\sum_{k=1}^{10} \int_0^1 f(k-1+x) dx =$
 1) 50 2) 10 3) 5 4) 0

27. $\int_0^{\infty} e^{-4x} \cdot \cos 3x dx =$
 1) $\frac{3}{25}$ 2) $\frac{4}{25}$ 3) $\frac{-1}{25}$ 4) $\frac{7}{25}$

PROBLEMS BASED ON INTEGRATION BY PARTS

28. $\int_1^2 \log x dx =$
 1) $2 \log 2 - 1$ 2) $\log 2 - 1$
 3) $2 \log 2 + 1$ 4) $2 \log 2 - 2$

29. $\int_2^e \left[\frac{1}{\log x} - \frac{1}{(\log x)^2} \right] dx =$
 1) $e - 2$ 2) $e + 2 \log_2 e$
 3) $e - 2 \log_2 e$ 4) $\log_2 e$

30. $\int_1^e \frac{\ln x}{x^2} dx =$
 1) $1 - \frac{2}{e}$ 2) $\frac{2}{e} - 1$ 3) $\frac{e-1}{e}$ 4) $1 + \frac{2}{e}$

31. $\int_0^{\pi^2/4} \cos \sqrt{x} dx =$
 1) 2 2) $\pi - 2$ 3) $\pi + 2$ 4) $(\pi/2) - 1$

32. $\int_0^1 (x-1)e^{-x} dx =$
 1) $-\frac{1}{e}$ 2) $\frac{1}{e}$ 3) e 4) $2e$

PROBLEMS BASED ON

$$\sqrt{(x-a)(b-x)}, \sqrt{\frac{x-a}{b-x}}$$

33. $\int_1^3 \frac{dx}{\sqrt{(x-1)(3-x)}} =$
 1) π 2) $-\pi$ 3) $\frac{\pi}{2}$ 4) 0

34. $\int_2^3 \frac{2-x}{\sqrt{5x-6-x^2}} dx =$
 1) $\pi/2$ 2) $-\pi/2$ 3) $-\pi/3$ 4) π

PROBLEMS BASED ON

$$f(a-x), f(a+b-x)$$

35. $\int_0^{\pi/2} \frac{\sqrt{\cot x}}{\sqrt{\tan x} + \sqrt{\cot x}} dx =$
 1) π 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{4}$

36. $\int_0^2 \frac{2x-2}{2x-x^2} dx =$
 1) 0 2) 1 3) -1 4) 4

37. $\int_0^{\pi/2} \frac{dx}{1+\tan^3 x} =$
 1) $\frac{\pi}{4}$ 2) $\frac{\pi}{2}$ 3) 1 4) $\log 2$

38. $\int_0^{\pi/2} \frac{1}{1+\sqrt[4]{\tan x}} dx =$
 1) $\pi/4$ 2) $\pi/2$ 3) 0 4) $\pi/3$

39. $\int_0^1 \log\left(\frac{1}{x}-1\right) dx =$
 1) 0 2) $-1/2$ 3) $1/3$ 4) $1/2$

40. The value of the integral $\int_3^6 \frac{\sqrt{x}}{\sqrt{9-x} + \sqrt{x}} dx$ is

1) $\frac{3}{2}$ 2) 2 3) 1 4) $\frac{1}{2}$

41. $\int_{\pi/6}^{\pi/3} \frac{\cos^3 x}{\sin^3 x + \cos^3 x} dx =$
 1) $\frac{\pi}{3}$ 2) $\frac{-\pi}{2}$ 3) $\frac{\pi}{6}$ 4) $\frac{\pi}{12}$

42. $\int_{\alpha}^{\frac{\pi}{2}-\alpha} \frac{\cot x}{\tan x + \cot x} dx =$
 1) $\frac{\pi}{4} - \alpha$ 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{8}$

PROBLEMS BASED ON $\int xf(x)$

43. If $I_1 = \int_1^2 x[\sqrt{x} + \sqrt{3-x}]dx$ and

$I_2 = \int_1^2 (\sqrt{x} + \sqrt{3-x})dx$ then $\frac{I_1}{I_2} =$

- 1) $\frac{1}{2}$ 2) $\frac{3}{2}$ 3) 2 4) 1

EVEN, ODD FUNCTIONS

44. $\int_{-\pi/2}^{\pi/2} e^{\sin^{-2}x} \cdot \sin^{2n+1}x dx =$

- 1) 0 2) $\pi/2$ 3) 1 4) $\pi/$

45. $\int_{-\pi/2}^{\pi/2} \log\left(\frac{2-\sin\theta}{2+\sin\theta}\right)d\theta =$

- 1) 0 2) 1 3) 2 4) -1

46. $\int_{\ln\lambda}^{\ln\left(\frac{1}{\lambda}\right)} \frac{f\left(\frac{x^2}{3}\right)(f(x)+f(-x))}{g(3x^2)(g(x)-g(-x))}dx =$

where $\lambda > 1$.

- 1) 0 2) 1 3) λ 4) $1/\lambda$

47. $\int_{-2}^2 (x^{11} \cos x + e^x)dx =$

- 1) $\sinh 2$ 2) $2 \sinh 2$
3) $\frac{3}{2} \sinh 2$ 4) $\frac{\sinh 2}{2}$

48. If $I = \int_{-a}^a (\alpha \sin^5 x + \beta \tan^3 x + \gamma \cos x) dx,$

where α, β, γ are constants, then the value depends on

- 1) γ, a 2) α, β, γ, a 3) α, β, a 4) β

49. The value of $\int_{-\pi/2}^{\pi/2} \sin\left\{\log\left(x + \sqrt{x^2+1}\right)\right\} dx$ is

- 1) 1 2) -1 3) 0 4) 2

50. The function $F(x) = \int_0^x \log(t + \sqrt{1+t^2})dt$ is

- 1) a periodic function 2) an even function
3) an odd function 4) even or odd

MODULUS FUNCTIONS

51. If $\int_1^4 |x-3|dx = 2A+B$ then A and B can be

- 1) $A = \frac{3}{2}, B = 4$ 2) $A = \frac{1}{2}, B = 1$
3) $A = 2, B = -\frac{3}{2}$ 4) $A = \frac{3}{2}, B = \frac{3}{2}$

52. Value of $\int_{-2}^3 |1-x^2|dx =$

- 1) $\frac{28}{3}$ 2) $\frac{14}{3}$ 3) $\frac{7}{3}$ 4) $\frac{1}{3}$

53. $\int_0^2 (|x| + |x-1|)dx =$

- 1) 1 2) -1 3) 2 4) 3

54. $\int_{e^{-1}}^{e^2} \left| \frac{\log x}{x} \right| dx =$

- 1) $\frac{3}{2}$ 2) $\frac{5}{2}$ 3) 3 4) 5

55. $\int_{1/2}^2 |\log_{10} x|dx =$

- 1) $\log_{10}(8/e)$ 2) $\frac{1}{2} \log_{10}(8/e)$
3) $\log_{10}(2/e)$ 4) $\log_e(3/e)$

56. $\int_0^{\pi} |\cos x - \sin x| dx =$

- 1) $4\sqrt{2}$ 2) $2\sqrt{2}$ 3) $4\sqrt{3}$ 4) $3\sqrt{2}$

57. If $f(t) = \int_{-t}^t \frac{e^{-|x|}}{2} dx$ then $\lim_{t \rightarrow \infty} f(t) =$

- 1) 1 2) $1/2$ 3) 0 4) -1

58. If $f(x) = \int_{-1}^x |t| dt$ then for any $x \geq 0, f(x) =$

- 1) $\frac{1}{2}(1-x^2)$ 2) $1-x^2$ 3) $\frac{1}{2}(1+x^2)$ 4) $1+x^2$

PERIODIC, STEP FUNCTIONS

59. If $I_1 = \int_0^{3\pi} f(\cos^2 x) dx$ and $I_2 = \int_0^{\pi} f(\cos^2 x) dx$ then

- 1) $I_1 = I_2$ 2) $I_1 = 2I_2$ 3) $I_1 = 5I_2$ 4) $I_1 = 3I_2$

60. $\int_0^{100} \sin(x - [x])\pi dx =$ (where $[]$ is g.i.f)

- 1) $\frac{100}{\pi}$ 2) $\frac{200}{\pi}$ 3) 100π 4) 200π

61. $\int_0^{1000} e^{x-[x]} dx =$ (where $[]$ is g.i.f)

- 1) $\frac{e^{1000}-1}{1000}$ 2) $\frac{e^{1000}-1}{e-1}$ 3) $1000(e-1)$ 4) $\frac{e-1}{1000}$

62. $\int_{-2}^{10} \text{sgn}(x - [x]) dx$ equals, where $[]$ denotes greatest integer function

- 1) -12 2) 10 3) 12 4) 8

63. $\int_0^4 2^{[x]} dx =$ (where $[x]$ is the greatest integer not more than x)

- 1) 10 2) 15 3) 20 4) 16

REDUCTION FORMULA

64. $\int_{-2\pi}^{2\pi} \cos^6 x dx =$

- 1) $\frac{4\pi}{5}$ 2) $\frac{5\pi}{4}$ 3) 15π 4) $\frac{\pi^2}{4}$

65. $\int_0^a \frac{x^5 dx}{\sqrt{a^2 - x^2}} =$

- 1) $\frac{a^5}{15}$ 2) $\frac{8a^5}{15}$ 3) $\frac{8a}{15}$ 4) $\frac{11a^2}{15}$

66. $\int_0^{2\pi} \sin^4 x \cdot \cos^6 x dx =$

- 1) $\frac{3\pi}{128}$ 2) $\frac{\pi}{128}$ 3) $\frac{5\pi}{128}$ 4) $\frac{7\pi}{128}$

67. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^4 x \cdot \cos^6 x dx =$

- 1) $\frac{3\pi}{128}$ 2) $\frac{3\pi}{256}$ 3) $\frac{5\pi}{128}$ 4) $\frac{7\pi}{128}$

68. $\int_0^{\frac{\pi}{6}} \cos^4 3\theta \sin^2 6\theta d\theta =$ (EAMCET -2014)

- 1) $\frac{5}{192}$ 2) $\frac{5\pi}{256}$ 3) $\frac{5\pi}{192}$ 4) $\frac{\pi}{96}$

PROBLEMS ON LEIBNITZ FORMULA

69. If $\int_{x^2}^{x^4} \sin \sqrt{t} dt$, then $f'(x)$ equals

- 1) $\sin x^2 - \sin x$ 2) $4x^3 \sin x^2 - 2x \sin x$
3) $x^2 \sin x^2 - x \sin x$ 4) $\sin x - \cos x$

70. $\lim_{x \rightarrow 0} \frac{x^4}{\int_0^{x^2} \tan^{-1} t dt} =$

- 1) 1 2) 2 3) 0 4) 3

71. $\lim_{x \rightarrow 0} \frac{\int_0^x \sin^3 t \cos t dt}{x^4} =$

- 1) 0.25 2) 2.5 3) 5.2 4) 0.52

PROBLEMS ON DEFINITE INTEGRATION AS LIMIT OF SUM

72. $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{\sqrt{4n^2 - r^2}} =$

- 1) $\pi/2$ 2) $\pi/3$ 3) $\pi/6$ 4) $\pi/5$

73. $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} \sqrt{\frac{n+r}{n-r}} =$

- 1) $\frac{\pi}{2}$ 2) 2π 3) $\frac{\pi}{2} - 1$ 4) $\frac{\pi}{2} + 1$

74. $\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{r=1}^n r \cdot e^{r/n} =$

- 1) e 2) 1 3) $2e$ 4) 2

75. $\lim_{n \rightarrow \infty} \left\{ \frac{1}{n} + \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{3n} \right\} =$

- 1) $\pi/4$ 2) $\pi/6$ 3) $\log 2$ 4) $\log 3$

$$76. \lim_{n \rightarrow \infty} \left[\frac{n+1}{n^2+1^2} + \frac{n+2}{n^2+2^2} + \dots + \frac{n+n}{n^2+n^2} \right] =$$

- 1) $\frac{\pi}{4} + \frac{1}{2} \log 2$ 2) $\frac{\pi}{4} - \frac{1}{2} \log 2$
3) $\frac{\pi}{2} + \frac{1}{2} \log 2$ 4) $\frac{\pi}{4} + \frac{1}{4} \log 2$

$$77. \lim_{n \rightarrow \infty} \left(\frac{1^m + 2^m + 3^m + \dots + n^m}{n^{m+1}} \right) =$$

- 1) $\frac{1}{m+1}$ 2) $\frac{1}{m+2}$ 3) $\frac{1}{m}$ 4) $\frac{1}{m+3}$

78 By the definition of the definite integral, the value of (EAMCET - 2014)

$$\lim_{n \rightarrow \infty} \left(\frac{1^4}{1^5 + n^5} + \frac{2^4}{2^5 + n^5} + \frac{3^4}{3^5 + n^5} + \dots + \frac{n^4}{n^5 + n^5} \right) \text{ is.}$$

- 1) $\frac{1}{5} \log 2$ 2) $\frac{1}{4} \log 2$ 3) $\frac{1}{4} \log 3$ 4) $\frac{1}{4} \log 2$

LEVEL-I (C.W.) - KEY

- | | | | | | |
|-------|-------|-------|-------|-------|-------|
| 1) 4 | 2) 2 | 3) 3 | 4) 2 | 5) 2 | 6) 2 |
| 7) 3 | 8) 3 | 9) 3 | 10) 1 | 11) 2 | 12) 2 |
| 13) 3 | 14) 1 | 15) 3 | 16) 3 | 17) 3 | 18) 2 |
| 19) 1 | 20) 4 | 21) 1 | 22) 3 | 23) 3 | 24) 3 |
| 25) 3 | 26) 3 | 27) 2 | 28) 1 | 29) 3 | 30) 1 |
| 31) 2 | 32) 1 | 33) 1 | 34) 2 | 35) 4 | 36) 1 |
| 37) 1 | 38) 1 | 39) 1 | 40) 1 | 41) 4 | 42) 1 |
| 43) 2 | 44) 1 | 45) 1 | 46) 1 | 47) 2 | 48) 1 |
| 49) 3 | 50) 2 | 51) 3 | 52) 1 | 53) 4 | 54) 2 |
| 55) 2 | 56) 2 | 57) 1 | 58) 3 | 59) 4 | 60) 2 |
| 61) 3 | 62) 3 | 63) 2 | 64) 2 | 65) 2 | 66) 1 |
| 67) 2 | 68) 3 | 69) 2 | 70) 2 | 71) 1 | 72) 3 |
| 73) 4 | 74) 2 | 75) 4 | 76) 1 | 77) 1 | 78) 1 |

LEVEL-I (C.W.) - HINTS

$$1. \quad GI = \frac{2}{5} \left[x^{\frac{5}{2}} \right]_0^4 = \frac{64}{5}$$

$$2. \quad GI = \frac{1}{\pi} \left[\tan^{-1} \left(\frac{x}{2} \right) \right]_0^2 = \frac{1}{\pi}$$

$$3. \quad \int_{\sqrt{2}}^x \frac{dx}{x\sqrt{x^2-1}} = \frac{\pi}{12} \Rightarrow [\sec^{-1} x]_{\sqrt{2}}^x = \frac{\pi}{12}$$

$$4. \quad \text{Use } \int \frac{1}{x^2+a^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}$$

$$5. \quad 2x+1=t$$

$$6. \quad \int_0^{\frac{\pi}{2}} \frac{\sin^2 x}{(1+\cos x)^2} dx = \int_0^{\frac{\pi}{2}} \tan^2(x/2) dx$$

$$7. \quad \sqrt{x}=t$$

$$8. \quad \text{using } \sqrt{1+\sin 2x} = \sin x + \cos x$$

$$9. \quad \int_0^{\pi/2} \sqrt{1+\sin x} dx = \int_0^{\pi/2} \left(\sin \frac{x}{2} + \cos \frac{x}{2} \right) dx$$

$$10. \quad \int_0^{\log 2} \cosh(2x) dx = \frac{1}{2} [\sinh 2x]_0^{\log 2}$$

$$11. \quad 1+\cos x = 2\cos^2(x/2)$$

12. Rationalise

$$13. \quad \int_0^{\infty} \left[\left(\frac{a}{c} \right)^x - \left(\frac{b}{c} \right)^x \right] dx$$

14. $f(x)$ is an odd function

$$15. \quad \int_0^1 \frac{dx}{e^x + e^{-x}} = \int_0^1 \frac{e^x dx}{(e^x)^2 + 1}, \text{ put } e^x = t$$

$$16. \quad \text{put } x^4 = t$$

$$17. \quad \sin x = t$$

$$18. \quad I = \int_0^{\pi/4} \frac{\sqrt{\tan x}}{\sin x \cos x} dx = \int_0^{\pi/4} \frac{\sqrt{\tan x}}{\tan x} \sec^2 x dx$$

$$= \int_0^{\pi/4} (\tan x)^{-\frac{1}{2}} \sec^2 x dx = \left[\frac{\tan^{\frac{1}{2}} x}{\frac{1}{2}} \right]_0^{\pi/4}$$

$$= 2(1-0) = 2$$

$$19. \quad \int_0^{\pi/2} \cos^5 x \cdot \sin 2x dx = 2 \int_0^{\pi/2} \cos^6 x \sin x dx$$

$$\text{put } \cos x = t$$

20. Put $x^x = t$
 21. split into two integrals
 22. Put $\frac{1}{x} = t$
 23. Put $1 + x^{4/5} = t$
 24. Put $\lambda x = t$, $\lambda dx = dt$, $\int_0^\infty e^{-\lambda x} x^{n-1} dx$

$$= \frac{1}{\lambda^n} \int_0^\infty e^{-x} x^{n-1} dx = \frac{\Gamma_n}{\lambda^n}$$

 25. Using $\int_{-1}^2 f(x) dx = \int_{-1}^4 f(x) dx - \int_2^4 f(x) dx$
 26. $\int_0^1 f(x) dx + \int_0^1 f(1+x) dx + \dots + \int_0^1 f(9+x) dx$

$$= \int_0^1 f(x) dx + \int_1^2 f(x) dx + \dots + \int_9^{10} f(x) dx$$

$$= \int_0^{10} f(x) dx = 5$$

 27. Using formula $\int_0^\infty e^{-ax} \cos bxdx = \frac{a}{a^2 + b^2}$
 28. $\int_1^2 \log x dx = [x \log x - x]_1^2 = 2 \log 2 - 1$
 29. $\log x = t$
 30. Using integration by parts
 31. $\sqrt{x} = t$
 32. Using the integration by parts
 33. $\int_a^b \frac{dx}{\sqrt{(x-a)(b-x)}} = \pi$
 34. $-\int_2^3 \sqrt{\frac{x-2}{3-x}} dx ; \int_a^b \sqrt{\frac{x-a}{b-x}} dx = \frac{\pi}{2}(b-a)$
 35. using formula $\int_0^{\frac{\pi}{2}} \frac{f(\cot x)}{f(\cot x) + f(\tan x)} dx = \frac{\pi}{4}$

36. Using formula $\int_0^a f(x) dx = \int_0^a f(a-x) dx$
 37. using formula $\int_0^{\frac{\pi}{2}} \frac{f(\cos x)}{f(\sin x) + f(\cos x)} dx = \frac{\pi}{4}$
 38. Using formula $\int_0^{2a} f(x) dx = 0$ If $f(2a-x) = -f(x)$
 39. Using formula $\int_0^{2a} f(x) dx = 0$ If $f(2a-x) = -f(x)$
 40. Using formula $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$
 41. $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$
 42. $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$
 43. $I_1 = \int_1^2 (3-x)[\sqrt{3-x} + \sqrt{x}] dx$

$$= \int_1^2 3(\sqrt{3-x} + \sqrt{x}) dx - \int_1^2 x(\sqrt{3-x} + \sqrt{x}) dx$$

 44. Given function is odd
 45. Given function is odd
 46. odd function
 47. $x^{11} \cos x$ is an odd function
 48. Solution : we have, I

$$\int_{-a}^a (\alpha \sin^5 x + \beta \tan^3 x + \gamma \cos x) dx$$

$$= \alpha \int_{-a}^a \sin^5 x dx + \beta \int_{-a}^a \tan^3 x dx + \gamma \int_{-a}^a \cos x dx$$

$$= \alpha \times 0 + \beta \times 0 + \gamma \times 2 \int_0^a \cos x dx$$

$$[\because \sin^5 x \text{ and } \tan^3 x \text{ are odd functions}]$$

$$\therefore I \text{ depends on } \gamma, a$$

 49. Since $\phi(x) = \sin \left\{ \log(x + \sqrt{x^2 + 1}) \right\}$

$$\phi(x) \text{ is an odd function.}$$

 Hence, $\int_{-\pi/2}^{\pi/2} \phi(x) dx = 0$

50. Solution we know that if $f(t)$ is an odd function,

then $F(x) = \int_0^x f(t) dt$ is an even function since

$\log(t + \sqrt{1+t^2})$ is odd, the given integral is even

$$51. \int_1^4 |x-3| dx = \int_1^3 |x-3| dx + \int_3^4 |x-3| dx$$

52. Given

$$\int_{-2}^{-1} (x^2-1) dx + \int_{-1}^1 -(x^2-1) dx + \int_1^3 (x^2-1) dx$$

$$53. \text{ Given } \int_0^2 x dx - \int_0^1 (x-1) dx + \int_1^2 (x-1) dx$$

$$54. \text{ Given } \int_{\frac{1}{e}}^1 -\frac{\log x}{x} dx + \int_1^{e^2} \frac{\log x}{x} dx$$

$$55. \int_{1/2}^2 |\log_{10} x| dx = \int_{1/2}^1 |\log_{10} x| dx + \int_1^2 \log_{10} x dx$$

$$= \log_{10} e \left[\int_{1/2}^1 -\log x dx + \int_1^2 \log x dx \right]$$

56. Given =

$$\int_0^{\frac{x}{a}} (\cos x - \sin x) dx + \int_{\frac{x}{a}}^0 -(\cos x - \sin x) dx$$

$$57. \text{ Even function } f(t) = 2 \int_0^t \frac{e^{-|x|}}{2} dx = \int_0^t e^{-x} dx$$

$$58. f(x) = \int_{-1}^0 -t dt + \int_0^x t dt$$

$$59. \text{ Let } g(x) = f(\cos^2 x)$$

$$g(x - n\pi) = f(\cos^2(x + n\pi))$$

$$\Rightarrow f(\cos^2 x) = g(x)$$

$$\int_0^{n\pi} g(x) dx = n \int_0^{\pi} g(x) dx \quad I_1 = 3I_2$$

$$60. \int_0^{100} \sin(x - [x]) \pi dx = 100 \int_0^1 \sin(x - [x]) \pi dx$$

$$61. \int_0^{1000} e^{x-[x]} x dx = 1000 \int_0^1 e^{x-[x]} dx = 1000 \int_0^1 e^x dx$$

$$62. x - [x] \geq 0; \int_{-2}^{10} \operatorname{sgn}(x - [x]) dx = \int_{-2}^{10} 1 dx$$

$$63. I = \int_0^1 2^0 dx + \int_1^2 2^1 dx + \int_2^3 2^2 dx + \int_3^4 2^3 dx$$

$$1 + 2 + 2^2 + 2^3 = 15$$

64. $\cos^6 x$ is even function and using reduction formula

65. Put $x = a \sin \theta$, and using reduction formula

$$66. \text{ Given } = 4 \int_0^{\frac{\pi}{2}} \sin^4 x \cos^6 x dx \text{ and using reduction formula}$$

67. Even function

$$68. \int_0^{\pi/6} \cos^4 3\theta \sin^2 6\theta d\theta =$$

$$\Rightarrow \frac{4}{3} \times \frac{1}{8} \times \frac{5}{6} \times \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2} = \frac{5\pi}{192}$$

69. Using Leibnitz rule

70. Using L's Hospital rule

$$\lim_{x \rightarrow 0} \frac{4x^3}{2x \tan^{-1}(x^2)} = \lim_{x \rightarrow 0} \frac{2x^2}{\tan^{-1}(x^2)} = 2$$

71. Using L's Hospital rule

$$\lim_{x \rightarrow 0} \frac{\sin^3 x \cos x}{4x^3} = \frac{1}{4} = 0.25$$

$$72. \text{ Given series } = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \frac{1}{\sqrt{4 - \left(\frac{r}{n}\right)^2}}$$

$$73. \text{ Given series } = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \sqrt{\frac{1 + \frac{r}{n}}{1 - \frac{r}{n}}}$$

$$74. \int_0^1 x e^x dx$$

75. Give series $= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{2n} \frac{1}{1 + \frac{r}{n}}$

76. Given series $= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \frac{1 + \frac{r}{n}}{1 + \left(\frac{r}{n}\right)^2}$

77. $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{2n} \frac{r^m}{n^{m+1}} = \int_0^1 x^m dx$

78. $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \frac{(r/n)^4}{(r/n)^5 + 1}$

$$\int_0^1 \frac{x^4}{x^5 + 1} dx = \left[\frac{1}{5} \log(x^5 + 1) \right]_0^1 = \frac{1}{5} \log 2$$



LEVEL-I (H.W)

PROBLEMS BASED ON BASIC FORMULAE

1. $\int_0^{\pi/4} \tan^2 x dx =$

1) $\frac{\pi}{4}$ 2) $1 - \frac{\pi}{4}$

3) $\frac{\pi}{2}$ 4) $1 + \frac{\pi}{4}$

2. $\int_0^{\pi} \sin mx \cdot \sin nx dx (m \neq n) =$

1) $\frac{\pi}{2}$ 2) 0 3) 1 4) π

3. $\int_0^{\infty} e^{-ax} dx (a > 0) =$

1) a 2) 0 3) $\frac{1}{a}$ 4) $-\frac{1}{a}$

4. $\int_0^a \frac{x-a}{x+a} dx =$

1. $a + 2a \log 2$ 2. $a - 2a \log 2$
3. $2a \log 2 - a$ 4. $2a \log 2$

5. $\int_1^2 \frac{dx}{\sqrt{1+x^2}} =$

1. $\log_e \left(\frac{2+\sqrt{5}}{\sqrt{2}+1} \right)$

2. $\log_e \left(\frac{\sqrt{2}+1}{2+\sqrt{5}} \right)$

3. $\log_e \left(\frac{2-\sqrt{5}}{\sqrt{2}-1} \right)$

4. 0

6. $\int_0^{\pi} \frac{\sin \theta + \cos \theta}{\sqrt{1 + \sin 2\theta}} d\theta =$

1) π

2) $\pi + \lambda$ and $\lambda > 0$

3) $\pi/2$

4) $\pi/3$

7. $\int_0^1 e^x \sinh x dx =$

1) $\frac{e^2 - 3}{4}$ 2) $\frac{e^2 + 3}{4}$ 3) $4(e^2 + 3)$ 4) $\frac{e^2 - 5}{4}$

8. $\int_0^a \sqrt{\frac{a+x}{a-x}} dx =$

1) $\frac{a}{2}(\pi + 2)$

2) $\frac{a}{2}(\pi - 2)$

3) $\frac{a}{3}(\pi + 2)$

4) $\frac{a}{2}(\pi + 3)$

PROBLEMS BASED ON SUBSTITUTIONS

9. $\int_0^{\pi/3} \frac{\cos x}{3 + 4 \sin x} dx =$

1) $\log \left(\frac{3 + 2\sqrt{3}}{3} \right)$

2) $\frac{1}{4} \log \left(\frac{3 + 2\sqrt{3}}{3} \right)$

3) $2 \log \left(\frac{3 + 2\sqrt{3}}{3} \right)$

4) $\frac{1}{2} \log \left(\frac{3 + 2\sqrt{3}}{2} \right)$

10. $\int_0^{\pi/4} \frac{\frac{1}{5} \sin^2 x}{\cos^2 x} dx =$

1) 0

2) $\frac{\pi}{4}$

3) 1

4) $2/3$

11. $\int_1^{e^3} \frac{dx}{x\sqrt{1+\ln x}} =$
 1) 2 2) $2\sqrt{2}$ 3) $\sqrt{2}$ 4) -2

12. $\int_0^1 \frac{1}{(x^2+1)^{\frac{3}{2}}} dx =$
 1) $\frac{1}{2}$ 2) $\frac{\sqrt{2}}{2}$ 3) 1 4) $\sqrt{2}$

13. $\int_{\pi/4}^{\pi/3} \operatorname{cosec} x \, d(\sin x) = \left(\text{for } 0 < x < \frac{\pi}{2} \right)$
 1) $\log 2$ 2) 0 3) $\log \left(\frac{\sin \frac{1}{2}}{\sin \frac{1}{\sqrt{2}}} \right)$ 4) $\frac{1}{2} \log \frac{3}{2}$

14. If $\int_0^{\infty} \frac{x^2 dx}{(x^2+a^2)(x^2+b^2)(x^2+c^2)} = \frac{\pi}{2(a+b)(b+c)(c+a)}$ then $\int_0^{\infty} \frac{dx}{(x^2+4)(x^2+9)} =$
 1) $\frac{\pi}{60}$ 2) $\frac{\pi}{20}$ 3) $\frac{\pi}{40}$ 4) $\frac{\pi}{80}$

15. If $\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$, then $\int_0^{\infty} e^{-ax^2} dx$, $a > 0$ is
 1) $\frac{\sqrt{\pi}}{2}$ 2) $\frac{\sqrt{\pi}}{2a}$ 3) $2 \frac{\sqrt{\pi}}{a}$ 4) $\frac{1}{2} \sqrt{\frac{\pi}{a}}$

16. If $f(x) = \begin{cases} e^{\cos x} \sin x, & \text{for } |x| \leq 2 \\ 2 & \text{otherwise} \end{cases}$ then $\int_{-2}^3 f(x) dx =$
 1) 0 2) 1 3) 2 4) 3

17. If $f: [-4, 5] \rightarrow R$ is defined by
 $f(x) = \begin{cases} \frac{|x|}{x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$, then $\int_{-4}^5 f(x) dx =$
 1) -1 2) 1 3) 0 4) $1/2$

18. $\int_0^{\infty} e^{-2x} \cdot \sin 5x dx =$
 1) $\frac{-2}{29}$ 2) $\frac{2}{29}$ 3) $\frac{5}{29}$ 4) $\frac{7}{25}$

19. $\int_{-4}^4 (2^x + 2^{-x})(3^x - 3^{-x}) dx =$
 1) 10 2) $16(\log 2)(\log 3)$
 3) $16 \log(2/3)$ 4) 0

20. $\int_0^{\infty} \frac{dx}{(x + \sqrt{x^2+1})^5} =$
 1) $1/24$ 2) $1/5$ 3) $5/24$ 4) $5/36$

PROBLEMS BASED ON INTEGRATION BY PARTS

21. $\int_0^1 \sin^{-1} \left(\frac{2x}{1+x^2} \right) dx =$
 1) $\frac{\pi}{4}$ 2) $\frac{\pi}{4} + \log 2$
 3) $\frac{\pi}{2} + \frac{1}{2} \log 2$ 4) $\frac{\pi}{2} - \log 2$

22. $\int_0^1 \frac{xe^x}{(x+1)^2} dx =$
 1) $\frac{e}{2}$ 2) $\frac{e-1}{2}$ 3) $\frac{e}{2} - 1$ 4) $\frac{e-3}{2}$

23. $\int_0^{\pi/2} e^x \left(\frac{1 + \sin x}{1 + \cos x} \right) dx =$
 1) $\pi/4$ 2) 0 3) $e^{\pi/2}$ 4) $e^{\pi/2} - 1$

24. $\int_0^1 (1+x) \cdot \log(1+x) dx =$
 1) $\log 4 - \frac{3}{4}$ 2) $\log 2 + \frac{3}{4}$
 3) $\frac{1}{2} \log 2 - \frac{3}{4}$ 4) $\log 4 + \frac{3}{4}$

PROBLEMS BASED ON

$$\sqrt{(x-a)(b-x)}, \sqrt{\frac{x-a}{b-x}}$$

25. $\int_0^1 \sqrt{x(1-x)} dx =$ (EAMCET - 1994)

- 1) $\pi/2$ 2) $\pi/4$ 3) $\pi/6$ 4) $\pi/8$

26. $\int_0^5 x(5-x)^{10} dx =$

1) $\frac{5^{12}}{132}$ 2) $\frac{5^{10}}{132}$ 3) 132.5^{12} 4) $\frac{132.5^{12}}{3}$

27. $\int_0^1 \frac{x}{(1-x)^4} dx =$

1) $16/3$ 2) $3/16$ 3) $-3/16$ 4) $-16/3$

28. $\int_0^{\pi/2} \frac{200 \sin x + 100 \cos x}{\sin x + \cos x} dx =$

1) 50π 2) 25π 3) 75π 4) 150π

29. $\int_0^{\pi/2} \frac{3^{\sec^3 x}}{3^{\sec^3 x} + 3^{\cos^3 x}} dx =$

1) 0 2) $\pi/2$ 3) $\pi/4$ 4) π

30. $\int_0^{\pi/2} \frac{f(\sec x) - f(\operatorname{cosec} x)}{1 + f(\sec x)f(\operatorname{cosec} x)} dx =$

1) 0 2) $\pi/2$ 3) $\pi/3$ 4) $\pi/4$

31. $\int_0^{\pi/2} \log \left(\frac{a + b \sin x}{a + b \cos x} \right) dx =$

1) 0 2) $\pi/4$ 3) $\pi/2$ 4) $\log \pi/2$

32. $\int_0^{2\pi} e^{\sin^2 nx} \cdot \tan nx \cdot dx =$ ($n \in N$)

1) 2 2) 1 3) π 4) 0

33. $\int_0^{\pi/2} \frac{\sin 8x \log(\cot x)}{\cos 2x} dx =$

1) 0 2) 1 3) $\frac{1}{2}$ 4) $\frac{\pi}{4}$

34. $\int_0^{\infty} \frac{\ln x}{1+x^2} dx$

1) π 2) 2π 3) 1 4) 0

PROBLEMS BASED ON $\int xf(x)$

35. $\int_0^{\pi} \frac{x \sin x dx}{1 + \cos^2 x} =$

1) $\frac{\pi^2}{4}$ 2) $-\frac{\pi^2}{2}$ 3) $\frac{\pi}{3}$ 4) $-\frac{\pi^2}{4}$

EVEN, ODD FUNCTIONS

36. $\int_{-a}^a x|x| dx =$

1) $\frac{a}{3}$ 2) $\frac{a^2}{3}$ 3) $\frac{a^2}{2}$ 4) 0

37. $\int_{-\pi/2}^{\pi/2} \sin |x| dx =$ (EAMCET - 2008)

1) 1 2) 2 3) -1 4) $1/2$

MODULUS FUNCTIONS

38. $\int_0^2 |x^2 + 2x - 3| dx =$

1) -1 2) 4 3) 2 4) 0

39. $\int_0^{\pi} \left(\cos x + \sqrt{\frac{1 + \cos 2x}{2}} \right) dx =$

1) 1 2) $1/2$ 3) 2 4) -1

40. $\int_0^{\pi/3} |\tan x - 1| dx =$

1) $\frac{\pi}{4} + \frac{1}{2}$ 2) $\frac{\pi}{6}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{4}$

41. $\int_{-1}^1 \frac{\sin x - x^2}{3 - |x|} dx =$

1) 0 2) $\int_0^1 \frac{\sin x}{3 - |x|} dx$

3) $\int_0^1 \frac{-2x^2}{3 - |x|} dx$ 4) $2 \int_0^1 \frac{\sin x - x^2}{3 - |x|} dx$

42. $I_1 = \int_0^{\pi/4} 2^{\sin x} dx, I_2 = \int_0^{\pi/4} 2^{\cos x} dx$ then

- 1) $I_1 = I_2$ 2) $I_1 > I_2$
3) $I_1 < I_2$ 4) Can't say

PERIODIC, STEP FUNCTIONS

43. $\int_{\pi}^{10\pi} |\sin x| dx =$

- 1) 20 2) 8 3) 10 4) 18

44. $\int_1^2 x^2 [x] dx =$ (where $[]$ is g.i.f)

- 1) 7/3 2) 8/3 3) 4/3 4) 5/3

45. $\int_1^4 \log [x] dx =$ (where $[]$ is g.i.f)

- 1) $\log 4$ 2) $\log 5$ 3) $\log 6$ 4) zero

REDUCTION FORMULA

46. $\int_{-a}^a \frac{x^4 dx}{\sqrt{a^2 - x^2}} =$

- 1) $\frac{3\pi a^4}{8}$ 2) $\frac{\pi a^4}{8}$ 3) $\frac{-\pi a^4}{8}$ 4) $\frac{5\pi a^4}{8}$

47. $\int_0^{\pi/2} \sin^6 x \cos^5 x dx =$

- 1) $\frac{8}{693}$ 2) $\frac{5}{693}$ 3) $\frac{4}{693}$ 4) $\frac{10}{693}$

48. If $I_n = \int_{\pi/4}^{\pi/2} (\tan \theta)^{-n} d\theta$ ($n > 1$) then $I_n + I_{n+2} =$

- 1) $\frac{1}{n+1}$ 2) $\frac{-1}{n+1}$ 3) $\frac{1}{n-1}$ 4) $\frac{-1}{n-1}$

49. If $I_n = \int_0^{\pi/4} \tan^n x dx$, then $\frac{1}{I_2 + I_4}, \frac{1}{I_3 + I_5}$

$\frac{1}{I_4 + I_6} \dots$ form (EAMCET-2010)

- 1) an A.P 2) a.G.P 3) a .H.P 4) A.G.P

PROBLEMS ON LEIBNITZ FORMULA

50. $\lim_{x \rightarrow 0} \left(\frac{\int_0^x \sin^2 t \cos t dt}{x^3} \right) =$

- 1) 1 2) 1/3 3) 1 4) 2

51. $\lim_{x \rightarrow 0} \left(\frac{\int_0^x \sin t dt}{\sin x^2} \right) =$

- 1) 1 2) 1/2 3) -1 4) 2

PROBLEMS ON DEFINITE INTEGRATION AS LIMIT OF SUM

52. $\lim_{n \rightarrow \infty} \left\{ \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{6n} \right\} =$

- 1) $\log 2$ 2) $\log 3$ 3) $\log 5$ 4) $\log 6$

53. $\lim_{n \rightarrow \infty} \left\{ \frac{1}{na} + \frac{1}{na+1} + \frac{1}{na+2} + \dots + \frac{1}{nb} \right\} =$

- 1) $\log \left(\frac{b}{a} \right)$ 2) $\log \left(\frac{a}{b} \right)$

- 3) $\log a$ 4) $\log b$

54. $\lim_{n \rightarrow \infty} \frac{1^9 + 2^9 + 3^9 + \dots + n^9}{n^{10}} =$

- 1) 1/2 2) 1/5
3) 1/10 4) 1/15

LEVEL-I (H.W.) - KEY

- | | | | | | |
|-------|-------|-------|-------|-------|-------|
| 1) 2 | 2) 2 | 3) 3 | 4) 2 | 5) 1 | 6) 3 |
| 7) 1 | 8) 1 | 9) 2 | 10) 4 | 11) 1 | 12) 2 |
| 13) 4 | 14) 1 | 15) 4 | 16) 3 | 17) 2 | 18) 3 |
| 19) 4 | 20) 3 | 21) 4 | 22) 3 | 23) 3 | 24) 1 |
| 25) 4 | 26) 1 | 27) 4 | 28) 3 | 29) 3 | 30) 1 |
| 31) 1 | 32) 4 | 33) 1 | 34) 4 | 35) 1 | 36) 4 |
| 37) 2 | 38) 2 | 39) 3 | 40) 2 | 41) 3 | 42) 3 |
| 43) 4 | 44) 1 | 45) 3 | 46) 1 | 47) 1 | 48) 1 |
| 49) 1 | 50) 2 | 51) 2 | 52) 4 | 53) 1 | 54) 3 |

LEVEL-I (H.W.) - HINTS

- $GI = \left[\tan x - x \right]_0^{\frac{\pi}{4}} = 1 - \frac{\pi}{4}$
- Use Transformations
- $GI = \left[\frac{e^{-ax}}{-a} \right]_0^{\infty} = \frac{1}{a} \sqrt{a^2 + b^2}$
- $\int_0^a \frac{x-a}{x+a} dx = \int_0^a \left(1 - \frac{2a}{x+a} \right) dx$
- $\int_1^2 \frac{dx}{\sqrt{1+x^2}} = \left[\sinh^{-1} x \right]_1^2$
 $= \left[\log(x + \sqrt{1+x^2}) \right]_1^2$
- $\int_0^{\pi} \frac{\sin \theta + \cos \theta}{\sqrt{1 + \sin 2\theta}} d\theta = \int_0^{\pi} \frac{\sin \theta + \cos \theta}{|\sin \theta + \cos \theta|} d\theta$
- $e^x \sinh x = e^x$
 $\left[\frac{e^x - e^{-x}}{2} \right] = \frac{e^{2x} - 1}{2}$
- $\int_0^a \sqrt{\frac{a+x}{a-x}} dx = \int_0^a \frac{a+x}{\sqrt{a^2 - x^2}} dx$
- Put $\sin x = t$
- $\int_0^{\frac{\pi}{4}} \frac{\sin^{\frac{1}{2}} x}{\cos^{\frac{5}{2}} x} dx$
 $= \int_0^{\frac{\pi}{4}} \tan^2 x \cdot \sec^2 x dx$
- $1 + \log x = t$
- Put $x = \tan \theta$
- $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \operatorname{cosec} x \cdot \cos x dx = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \cot x dx$
- $\int_0^{\infty} \frac{dx}{(x^2+4)(x^2+9)} = \int_0^{\infty} \frac{x^2}{(x^2+0)(x^2+4)(x^2+9)} dx$
- Put $t = \sqrt{ax}$

- Solution we have, $\int_{-2}^3 f(x) dx$
 $= \int_{-2}^2 e^{\cos x} \sin x dx + \int_2^3 2 dx$
 $= 0 + 2(3-2) \left[\begin{array}{l} \because g(x) = e^{\cos x} \sin x \text{ is an odd function} \\ \therefore \int_{-2}^2 g(x) dx = 0 \end{array} \right]$
- $I = \int_{-4}^0 \frac{-x}{x} dx + \int_0^5 \frac{x}{x} dx = (-x)_{-4}^0 + (x)_0^5 = -4 + 5 = 1$
- Using formula $\int_0^{\infty} e^{-ax} \sin bx dx = \frac{b}{a^2 + b^2}$
- $(2^x + 2^{-x})(3^x - 3^{-x})$ is an odd function
- Using formula $\int_0^{\infty} \frac{dx}{(x + \sqrt{x^2 + 1})^n} = \frac{n}{n^2 - 1}$
- Put $x = \tan \theta \Rightarrow GI = 2 \int_0^1 \tan^{-1} x dx$
- $\int e^x [f(x) + f'(x)] dx = e^x f(x)$
- Use $\int e^x (f(x) + f'(x)) dx = e^x f(x)$
- Using the integration by parts
- $\int_b^a \sqrt{(x-a)(b-x)} dx = \frac{\pi}{8} (b-a)^2$
- Using formula $\int_0^a f(x) dx = \int_0^a f(a-x) dx$
- Using formula $\int_0^a f(x) dx = \int_0^a f(a-x) dx$
- $\int_0^{\frac{\pi}{2}} \frac{a \cos x + b \sin x}{\cos x + \sin x} dx = \frac{\pi}{4} (a+b)$
- Using formula $\int_0^{\frac{\pi}{2}} \frac{f(\sec x)}{f(\sec x) + f(\operatorname{cosec} x)} dx = \frac{\pi}{4}$

30. Using formula $\int_0^a f(x)dx = \int_0^a f(a-x)dx$

31. $\int_0^{\frac{\pi}{2}} \frac{f(\cos x)}{f(\sin x) + f(\cos x)} dx = \frac{\pi}{4}$

32. Using $\int_0^a f(x)dx = \int_0^a f(a-x)dx$

33. Using formula $\int_0^a f(x)dx = 0$ If $f(a-x) = -f(x)$

34. Put $x = \tan \theta$

35. Using formula $\int_0^a f(x)dx = \int_0^a f(a-x)dx$

and put $\cos x = t$

36. Given function is odd

37. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin |x| dx = 2 \int_0^{\frac{\pi}{2}} \sin |x| dx = 2 \int_0^{\frac{\pi}{2}} \sin x dx$

38. $\int_0^2 (x^2 + 2x - 3) dx =$

$\int_0^1 (x^2 + 2x - 3) dx + \int_1^2 (x^2 + 2x + 3) dx$

39. Given $\int_0^{\pi} (\cos x + |\cos x|) dx$

$= \int_0^{\frac{\pi}{2}} \cos x dx + \int_{\frac{\pi}{2}}^{\pi} \cos x dx + \int_{\frac{\pi}{2}}^{\pi} -\cos x dx$

40. Given $= \int_0^{\frac{\pi}{4}} -(\tan x - 1) dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} (\tan x - 1) dx$

41. $\int_{-1}^4 \frac{\sin x - x^2}{3 - |x|} dx = \int_{-1}^4 3 - |x| dx - \int_{-1}^4 \frac{x^2}{3 - |x|} dx$

$= 0 - \int_0^4 \frac{2x^2}{3 - |x|} dx$

($\because$ first integrand is an odd function and second is an even function)

42. $\cos x > \sin x \forall x \in (0, \pi/4)$

43. $\int_{mT}^{nT} f(x) dx = (n-m) \int_0^T f(x) dx$

44. $1 < x < 2 \Rightarrow [x] = 1$

45. Given $= \int_1^2 \log [x] dx + \int_2^3 \log [x] dx + \int_3^4 \log [x] dx$

46. Put $x = a \sin \theta$, and using reduction formula

47. $\int_0^{\frac{\pi}{2}} \sin^m x \cos^n x dx$
 $= \frac{(m-1)(m-3)\dots\dots(n-1)(n-3)\dots\dots}{(m+n)(m+n-2)\dots\dots} k$

48. $I_n + I_{n+2} = \frac{1}{n+1}$ if $I_n = \int_0^{\pi/4} \tan^n x dx$

49. $I_n + I_{n+2} = \frac{1}{n+1}$ if $I_n = \int_0^{\pi/4} \tan^n x dx$

50. Using L's Hospital rule $\lim_{x \rightarrow 0} \frac{\sin^2 x \cos x}{3x^2} = \frac{1}{3}$

$\lim_{x \rightarrow 0} \frac{\sin x}{2x \cos(x^2)} = \frac{1}{2}$

51. Using L's Hospital rule

52. Given series $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{5n} \frac{1}{1 + \frac{r}{n}}$

53. $\lim_{n \rightarrow \infty} \sum_{r=0}^{\lfloor b-a \rfloor n} \frac{1}{na+r} = \int_0^{b-a} \frac{1}{a+x} dx$

54. Given series $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n} \right)^9$